

Name: _____

Multiple Integrals Practice

(1.) Consider the following expressions.

$$I = \iint_R f(x, y) \, dA \quad \text{and} \quad J = \iiint_T f(x, y, z) \, dV \quad (1)$$

(i.) In words, what does I represent?

(i.) In words, what does J represent? (The 4D meaning)

(ii.) As a mathematical expression, what is the definition of I

(iv.) If $f(x, y, z)$ is a mass density function on the region T , what does J represent?

(2.) Consider the following double integral

$$\iint_R 5x^3 \cos y^3 \, dA \tag{2}$$

where R is bounded by $y = 2$ and $y = \frac{1}{4}x^2$.

(i.) Evaluate the double integral.

(ii.) In words, interpret your previous answer.

(3.) Consider the following double integral

$$\iint_R e^{-x^2-y^2} \, dA \tag{3}$$

where R is the region between $x^2 + y^2 = 2x$ and $x^2 + y^2 = 8x$ and $y = 0$.

(i.) Evaluate the double integral.

(ii.) Using calculus find the area of R .

(4.) Consider the following triple integral.

$$\iiint_T \sqrt{x^2 + y^2} \, dV \quad (4)$$

Where T is the region which lies inside the cylinder $x^2 + y^2 = 16$, and between planes $z = 1$ and $z = y + 1$

(i.) Evaluate the triple integral.

(5) Suppose $\rho(x, y, z) = 15z$ is the mass density function over the 3-dimensional region T which is bounded by $2x + y + z = 4$, $4x + 4y + 2z = 20$, $z = 2y^2$, and $z = \sqrt{4y}$.

(i.) Find the mass of the T .